

Vol 4 Chapter 2 — Gravity as Cumulative Eigentone Effect

Keeper (author pass)

2026-05-23 Saturday

Chapter 2 — Gravity as Cumulative Eigentone Effect

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Chapter 1 derived Newton's G from the substrate's Bergman geometry with the substrate-mechanism reading that gravity is the substrate's *residual* curvature after the strong, electromagnetic, and weak sectors have completed their per-tick commitments. This chapter develops the residual-curvature picture into a full substrate-derivation of the Einstein field equations.

2.1 The eigentone picture

Lyra T2106 (April 2026) frames gravity as the substrate's cumulative eigentone effect — the accumulated curvature that integrates across many substrate ticks at long-distance Scale 3 (Volume 0 Chapter 3 §3.4). Each substrate tick produces commitments in the strong, electromagnetic, and weak sectors; what remains uncommitted is the substrate's residual Bergman curvature, which at large scales integrates to produce the gravitational field.

The substrate's six-dimensional internal Bergman computation — the K-type structure on $H^2(D_{IV}^5)$ — has a four-dimensional residual projection that is exactly the gravitational sector. The integration runs across the substrate's complete K-type spectrum, with the gauge sectors absorbing their respective contributions and gravity inheriting what is left.

2.2 Einstein equations from substrate commitment

The Einstein field equation $G_{\mu\nu} = 8\pi GT_{\mu\nu}$ has, in BST, a substrate-mechanical derivation via the substrate's commitment-cycle structure. The lefthand-side Ein-

stein tensor $G_{\mu\nu}$ is the substrate’s residual Bergman curvature; the righthand-side stress-energy tensor $T_{\mu\nu}$ is the substrate’s matter content; the coupling $8\pi G$ inherits the four-round-trip suppression we derived in Chapter 1.

Lyra’s `BST_EinsteinEquations_FromCommitment.md` paper-grade note (April 2026) develops the substrate-derivation in detail. The substrate-mechanism reading is that Einstein’s equations are not postulated — they are the substrate’s residual-curvature-equals-matter-content closure condition.

2.3 Why no quantum gravity problem

Standard physics has struggled for decades to quantize gravity. The fundamental issue: General relativity treats the metric as the dynamical field; quantization of the metric runs into renormalization difficulties at the Planck scale.

In BST, there is no quantum gravity problem because gravity is *not a fundamental sector*. The metric is the substrate’s residual curvature; quantizing the substrate’s commitment cycle automatically quantizes the residual. The substrate’s Reed–Solomon discretization on $GF(128)^k$ is the substrate-side ultraviolet completion; there is no Planck-scale divergence to renormalize away.

This recasts quantum gravity as a *substrate-mechanism* problem rather than a fundamental-field-theory problem. The substrate is already quantum; gravity emerges from the substrate’s commitment computation; the standard quantum-gravity puzzle dissolves.

2.4 What comes next

Chapter 3 examines the BST-SR / BST-GR boundary — where the substrate’s special-relativistic behavior gives way to general-relativistic behavior. Chapter 4 derives the cosmological constant (already rewritten as the second Vol 4 recruiter chapter).

Where to look this up: T2106 (gravity as cumulative eigentone). `notes/BST_EinsteinEquations_FromCommitment`. (substrate-derivation of Einstein equations). T1296 (Newton’s G , Chapter 1). For standard GR/quantum gravity: Misner, Thorne, Wheeler, *Gravitation*; Rovelli, *Quantum Gravity*.